

De Novo Peptide Design using Evolutionary Algorithms and Molecular Docking

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Abstract—This project explores the application of Evolutionary Algorithms (EA) for the *de novo* design of peptide ligands targeting specific protein receptors. The problem of finding a peptide sequence with high binding affinity is combinatorial and computationally expensive. We propose a Genetic Algorithm (GA) implemented using the `inspyred` library, where the fitness evaluation is carried out with a physics-based docking simulation run on `AutoDock Vina`. The pipeline integrates `OpenBabel` for chemical file conversion and customized genetic operators (mutation, crossover) designed for amino acid sequences. Experimental results demonstrate the algorithm’s ability to evolve peptides with progressively lower binding energies (higher affinity) over generations.

Index Terms—Evolutionary Algorithms, Molecular Docking, De Novo Design, Genetic Algorithms.

I. INTRODUCTION

THE design of high-affinity peptide binders for protein targets is a central challenge in computational biology and drug discovery. Traditional methods often rely on high-throughput screening, which is costly and time-consuming. Moreover, in addition to a high binding affinity, to be commercialized, a drug must be characterized by a low hydrophobicity index, so as to ensure optimal solvation in water and absorption in the host.

Computational approaches, specifically Molecular Docking, allow for the prediction of the binding orientation and affinity of a ligand to a receptor. However, the search space for peptide sequences is vast (20^N for a peptide of length N). To navigate this space efficiently, Evolutionary Computation provides valid tools that allow for the optimization of peptide sequences.

In this work, we implement an Evolutionary Algorithm that evolves a population of peptide sequences. Given as input the protein of interest, with a custom binding box built around the enzyme pocket, a randomly initialized population of peptide sequences is generated and evolved to optimize the binding affinity.

The random generation of sequences enables unbiased research in the solution space without the constraint of using previously known or existing peptides. Thus, new and innovative ligands can be generated with the focus entirely shifted to specificity and applicability, without the need to rely on present databases. The physics-based docking method employed by `AutoDock Vina`, which exploits the computation of short-range Van der Waals interactions, hydrogen bonding, hydrophobic interaction, and steric hindrance to output a scoring function in the form of binding energy, allows for meaningful results even in the presence of rather exotic peptides.

Through this method, we were able to generate peptides with binding affinities up to -11.2 kcal/mol while still maintaining acceptable hydrophobicity indices.

II. METHODOLOGY

A. Data and Code Availability

The source code used for the evolutionary algorithm, along with the `Vina` docking scripts and result logs, is openly available at: https://github.com/ale-0x/bioAI_project/tree/v1.2.1.

B. System Architecture

The pipeline is implemented in Python and orchestrates several bioinformatics tools:

- **Insyred [1]:** Framework for the Evolutionary Algorithm logic.
- **AutoDock Vina:** External engine for docking simulations and scoring.
- **OpenBabel [2], RDKit [3]:** Tools for preprocessing of the receptor.
- **PyMOL API [4]:** Used to identify the binding pocket centroid coordinates.

C. Preprocessing

The receptor is prepared for docking using `RDKit` / `OpenBabel`. The structure is optimized (embedding) and protonated at physiological pH and converted to `PDBQT` format suitable for `Vina`.

D. Problem Representation

Each individual in the population is represented as a list of one-letter codes that represent available amino acids (e.g., `['A', 'C', 'D', ...]`). The standard 20 amino acids are used as the alphabet for the genetic code, and the sequence length is fixed at initialization.

E. Evolutionary Operators

To maintain genetic diversity and explore the landscape, we implemented domain-specific operators:

1) *Crossover*: We use a standard *One-Point Crossover*. Two parent peptides are split at a random index, and their tails are swapped to create two offspring.

2) *Mutation*: At each position, a point mutation is applied with a certain probability to exchange the current amino acid with a random one. To enhance exploitation as the algorithm proceeds, the mutation probability is linearly reduced with the generations from an `INITIAL_MUTATION_RATE` to a `FINAL_MUTATION_RATE`. At each position, if the mutation occurs, the probability distribution for the sampling of the new amino acid is taken from the **BLOSUM62** Substitution Matrix [5], which contains substitution probabilities derived from experimental observations.

F. Fitness Function

The fitness function is the core of the evaluation. For a given candidate peptide, its fitness is defined as:

$$f(x) = \Delta G + \frac{\lambda}{n} \sum_i^{len(x)} h_i$$

where ΔG is the free binding energy obtained by performing a real-time docking simulation [6], with an additional term reflecting the hydrophobicity value at pH7 (h) computed on average across the amino acid sequence, penalizing less soluble sequences.

AutoDock Vina is executed with a previously defined search box centered on the receptor's active site. The returned docking score (lowest binding energy in kcal/mol) is extracted. Favorable binding is found when the free energy difference between the bound/unbound configurations is negative, meaning that the entropy of the bound state is **lower** than that of the one with separated individual parts. As a result, that configuration is thermodynamically favoured, and binding is spontaneous. Therefore, the EA is configured to **minimize** the fitness value.

G. Evolutionary Strategy

We utilized a Genetic Algorithm to evolve the sequence population. The algorithm implements a 1:1 spawning from the parent population to the offspring. In this way, each parent generates exactly one new sequence. As a replacement technique, a $(\lambda + \lambda)$ strategy was implemented, creating the next generation by collecting the best individuals among combined parents and offspring. Due to the high dimension of the research space, the aim of the algorithm is to progressively shift towards exploitation by reducing the mutation rate with each generation.

III. EXPERIMENTAL SETUP

The experiments were conducted on a SLURM-based cluster [7]. The parameters used for the evolution are detailed in Table I.

IV. RESULTS

Due to limitations on the cluster access, only 16 generations were achieved. Despite the limited number, it is possible to appreciate the evolution of the scoring function from Fig. 1 and Fig. 2, where it is possible to observe the decrease of the fitness towards an optimal area of the solution space.

TABLE I
GENETIC ALGORITHM AND DOCKING CONFIGURATION

Parameter	Value
Peptide Length	6
Population Size	50
Max Generations	50
Initial Mutation Rate	0.60
Final Mutation Rate	0.15
Crossover Rate	1.0
Hydrophobicity Weight	0.015
Vina Exhaustiveness	8
Search Space (Box Size)	$32 \times 32 \times 32 \text{ \AA}$

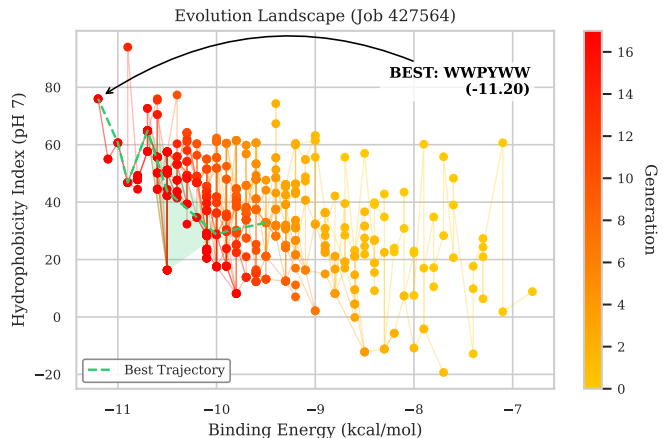


Fig. 1. **Fitness vs. Hydrophobicity Evolution.** Scatter plot showing the distribution of candidates across generations (color-coded). The x-axis represents Vina binding energy, while the y-axis shows the hydrophobicity index at pH 7. A distinct convergence trend is visible: as generations progress, individuals minimize binding energy, moving towards the high-affinity region on the left. Green dashed line highlights the step-by-step improvement of the best candidate throughout the simulation.

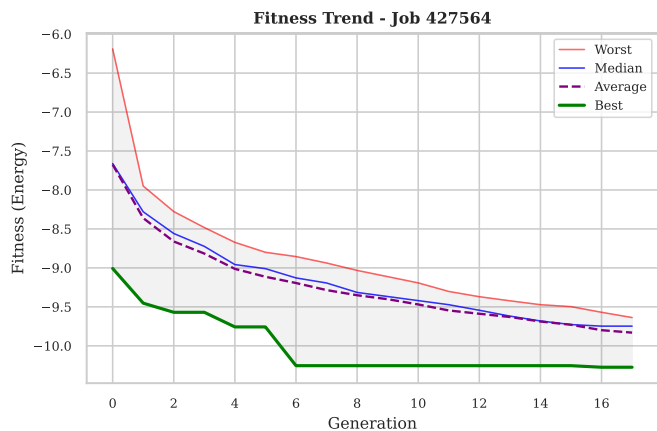


Fig. 2. **Fitness Trends by Generation.** Evolution of population metrics (Best, Average, Median, Worst). A rapid initial convergence is observed, with the best fitness (green line) reaching a stable minimum of -10.05 kcal/mol as early as generation 6, maintaining it until the end of the simulation.

A. Discussion

The best peptide found was sequence "WWPYWW" with a binding affinity of **-11.2 kcal/mol** and an average hydrophobicity index of 1.14.

The candidate was inspected manually with PyMol's viewer (Fig. 3) and checked for similarity with the original ligand (Fig. 4). Furthermore, the presence of polar contacts was highlighted, supporting the viability of the candidate for screening as a binder for the pocket of interest (Fig. 5).

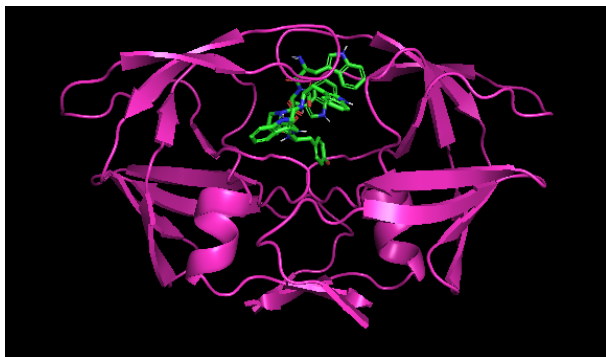


Fig. 3. **Structural Binding Mode.** Docking visualization for the sequence WWPYWW. The peptide (green) entirely occupies the binding cavity of the receptor (magenta), demonstrating high geometric complementarity.



Fig. 4. **Reference Superposition.** Comparison between the evolved peptide (green) and the original ligand 3TL (yellow). The close conformational alignment (Right) demonstrates that the candidate reproduces the binding mode of the benchmark ligand, adhering to the active site's steric constraints.

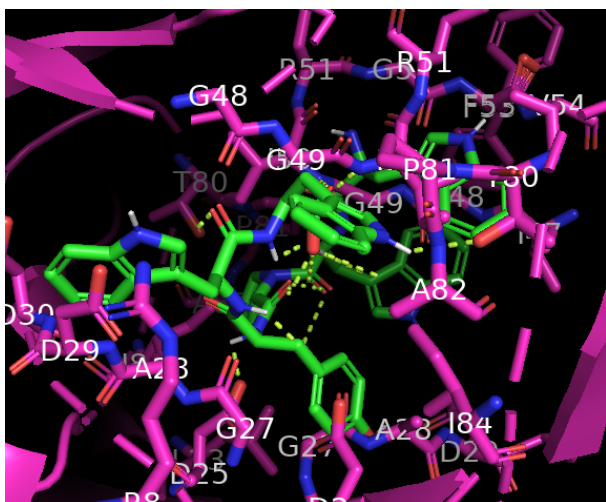


Fig. 5. **Polar Contacts Analysis.** Close-up visualization of the ligand-receptor interface. Dashed lines mark polar interactions with residues within a 4 Å cutoff, revealing the chemical determinants of the binding affinity.

At the population level, the best candidates generated across the whole experiment were inspected for similarity and the

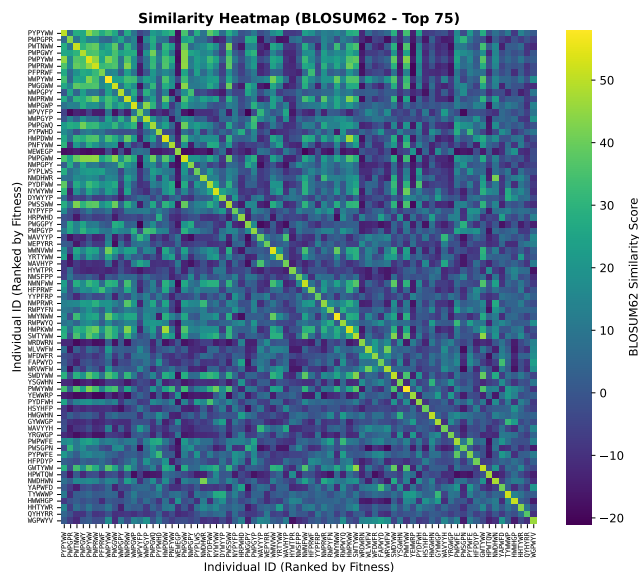


Fig. 6. **BLOSUM62 Similarity Matrix.** Comparative analysis of the 75 highest-fitness sequences. Pairwise similarity relationships are shown: warm colors (yellow/green) identify clusters of similar sequences, highlighting how the best solutions tend to cluster around specific conserved motifs, while maintaining some heterogeneity (blue areas).

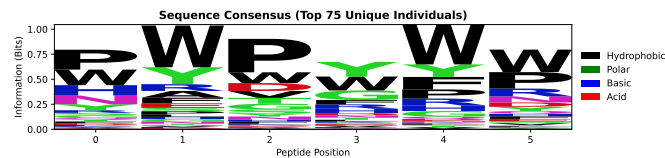


Fig. 7. **Sequence Motif of the Top-Ranked Individuals.** Sequence Logo representation of the 75 sequences with the best fitness. The Y-axis indicates the entropy of the information (bits): higher letters correspond to conserved amino acids, while positions with stacked letters indicate greater residual variability.

presence of a shared pattern. This analysis allows for selecting more than a single candidate for a putative screening panel, and evaluating simultaneously similar viable solutions. The results are shown in a similarity heatmap and with a pattern logotype (Fig. 6 and Fig. 7).

During the run, an unexpected trend was observed in the hydrophobicity index, which was observed to rise along the generations. To investigate the trend, the experiment was repeated with the same parameters, changing only the hydrophobicity weight to zero. As a result, we observed a great increase in the index value associated with the increasing of the binding affinity. (Fig. S1 and Fig. S2)

An explanation for the high values of the hydrophobicity index, regardless of our attempt to penalize it, can be derived from the characteristics of the binding pocket of the target protein used in the evaluation. By looking at the original ligand 3TL, it is possible to observe a particularly high presence of hydrophobic moieties, disclosing how the binding pocket prefers non-soluble sequences. For these reasons, seems plausible that, despite the penalization, the algorithms persist in increasing the binding affinity by decreasing the solubility.

It remains that, by tuning the associated weight, it is possible

to drive the research towards the desired characteristic and force the system to explore more thoroughly the solution space, eventually finding different minima.

V. CONCLUSION

This project successfully demonstrated an automated pipeline for peptide drug design. By combining *inspyred* with AutoDock Vina, we created a system capable of evolving high-affinity binders with little human intervention. By starting from a random population, we were able to obtain sequences with an affinity of -11.2 kcal/mol, comparable with values observed in strong crystal interactions. Despite the low number of generations evaluated, it was possible to derive a smooth decreasing trend during the minimization of the fitness, proving the correct function of the algorithm and the increment in binding affinity through the evolution process.

Future work will focus on introducing flexibility into receptor side chains and optimizing the hydrophobicity penalization effect.

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VI. SUPPLEMENTARY MATERIALS



Fig. S1. **Statistics (No Hydrophobicity).** The metrics show a convergence toward lower binding energies than in the constrained case. Without the hydrophobic penalty, selection pushes the population toward tighter theoretical global minima, ignoring *drug-likeness* criteria.

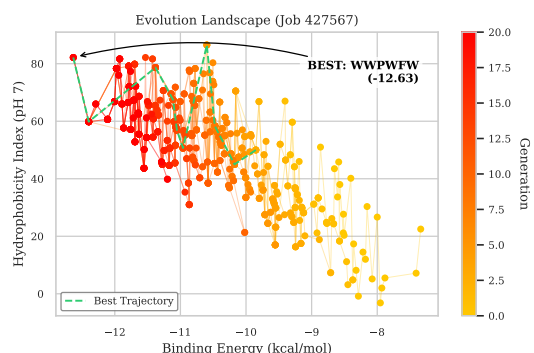


Fig. S2. **Intrinsic Binding Preference.** In the absence of hydrophobicity calculations, hydrophobic sequences are naturally favored; this trend likely reflects the chemical properties of the original ligand in the crystal structure.

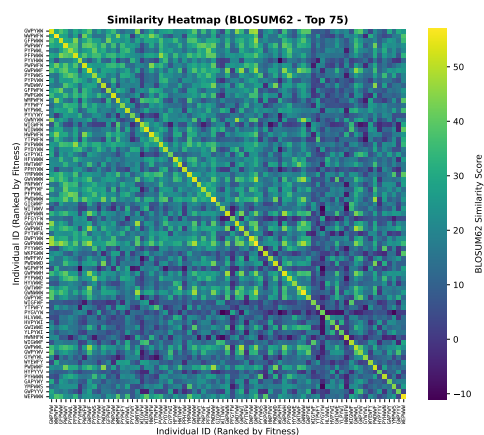


Fig. S3. **Unconstrained Diversity Loss.** Similarity heatmap (BLOSUM62) for the experiment without hydrophobic penalty. The prevalence of high scores (warm colors) off the diagonal highlights a strong convergence toward homologous solutions, confirming that the absence of solubility constraints drastically reduces the genetic variability of the elite population.

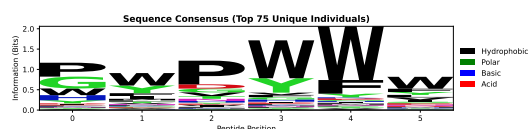


Fig. S4. **Unconstrained Motif Fixation.** Consensus analysis for the top 75 (no hydrophobicity). Zero entropy values (maximum letter height) indicate that the population is nearly identical, dominated exclusively by hydrophobic amino acids (W, P) optimized for binding affinity alone.

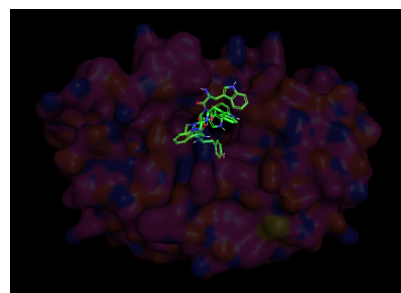


Fig. S5. **Hydrophobic Pocket Occupancy.** Surface view visualization. The ligand WWPYWW is clearly positioned within the receptor's hydrophobic cleft, demonstrating excellent steric complementarity and a deep fit into the binding pocket.